PROGRAMMING IN JAVA

Assignment VII

TYPE OF QUESTION: MCQ

Number of questions: 10 Total marks**: 10 × 1 = 10**

QUESTION 1:

What will be the output of the following Java program?

```
import java.io.*;
class ReadFile {
    public static void main(String[] args) throws IOException {
        FileReader fr = new FileReader("NPTEL.txt");
        BufferedReader br = new BufferedReader(fr);
        System.out.println(br.readLine());
        br.close();
    }
}
```

Assume `NPTEL.txt` contains:

```
This is Programming in Java online course.
```

a. **Hello, World!**

b. **This is Programming in Java online course.**

c. **IOException**

d. **null**

Correct Answer: b

b. **This is Programming in Java online course.**

Detailed Solution:

`BufferedReader` to read the first line from the file `NPTEL.txt`. Since the file contains “This is Programming in Java online course.”, it is printed.

QUESTION 2:

Complete the following snippet with the required code.

```
File file = new File("file.txt");
if (__________________) // Fill in the blanks
{
    System.out.println("File exists.");
} else
{
    System.out.println("File does not exist.");
}
```

a. **file.exists()**

b. **file.isFile()**

c. **file.fileExists()**

d. **file.isAvailable()**

Correct Answer: a

a. **file.exists()**

Detailed Solution:

The `exists()` method in the `File` class checks if a file or directory exists in the specified path.

QUESTION 3:

What is the output of the following Java program?

```
class Main {

  public static void main(String args[]) {
    final int i;
    i = 20;
    i = 30;
    System.out.println(i);
  }
}
```

a. **30**

b. **Compiler Error**

c. **Garbage value**

d. **0**

Correct Answer: b

b. **Compiler Error**

Detailed Solution:

i is assigned a value twice. Final variables can be assigned values only one. Following is the compiler error “Main.java:5: error: variable i might already have been assigned”.

QUESTION 4:

Which of these values is returned by read() method when the end of file (EOF) is encountered?

a. **0**

b. **1**

c. **-1**

d. **Null**

Correct Answer: c

c. **-1**

Detailed Solution:

Each time read() is called, it reads a single byte from the file and returns the byte as an integer value. read() returns -1 when the end of the file is encountered.

QUESTION 5:

What will be the output of the following Java program?

```
public class Calculator {
    int num = 100;

    public void calc(int num) {
        this.num = num * 10;
    }

    public void printNum() {
        System.out.println(num);
    }

    public static void main(String[] args) {
        Calculator obj = new Calculator();
        obj.calc(2);
        obj.printNum();
    }
}
```

a. **20**

b. **100**

c. **1000**

d. **2**

Correct Answer: a

a. **20**

Detailed Solution:

Here the class instance variable name(num) is same as calc() method local variable name(num). So for referencing class instance variable from calc() method, this keyword is used. So in statement this.num = num * 10, num represents local variable of the method whose value is 2 and this.num represents class instance variable whose initial value is 100. Now in printNum() method, as it has no local variable whose name is same as class instance variable, so we can directly use num to reference instance variable, although this.num can be used.

QUESTION 6:

What is the primary difference between Byte Streams and Character Streams in Java?

a. **Byte Streams handle characters, while Character Streams handle bytes.**

b. **Byte Streams are used for binary data, while Character Streams are used for image data.**

c. **Character Streams are faster than Byte Streams in all cases.**

d. **Byte Streams are used for binary data, while Character Streams are used for text data.**

Correct Answer: d

d. **Byte Streams are used for binary data, while Character Streams are used for text data.**

Detailed Solution:

Byte Streams: Handle raw binary data like images or files (InputStream, OutputStream).
Character Streams: Handle text data and support encoding like Unicode (Reader, Writer).
Character Streams are better for text processing, especially when handling international characters.
For a detailed understanding of Java I/O streams, refer to the book *Joy with Java*.

QUESTION 7:

What is the primary purpose of thread synchronization in Java?

a. **To allow multiple threads to execute a method at the same time**

b. **To ensure thread execution follows a specific order**

c. **To prevent rare conditions and ensure data in-consistency**

d. **To prevent race conditions and ensure data consistency**

Correct Answer: d

d. **To prevent race conditions and ensure data consistency**

Detailed Solution:

Thread synchronization is used to control the access of multiple threads to shared resources, ensuring data consistency and preventing race conditions. This is typically done using synchronized methods or blocks.
For more details on thread synchronization, refer to the book *Joy with Java*.

QUESTION 8:

Which stream does Java application uses to only READ data from a source, it may be a file, an array, peripheral device or socket?

a. **InputStream**

b. **OutputStream**

c. **Input/OutputStream**

d. **DataOutputStream**

Correct Answer: a

a. **InputStream**

Detailed Solution:

`InputStream` is used to read data from any source in Java.

QUESTION 9:

Which class in Java is used to create a new directory?

a. **FileReader**

b. **FileWriter**

c. **File**

d. **Directory**

Correct Answer: c

c. **File**

Detailed Solution:

The File class is used to create directories and manipulate files.

QUESTION 10:

What is the primary purpose of input streams in Java?

a. **To write data to a file.**

b. **To read data from a file.**

c. **To delete a file.**

d. **To create directories.**

Correct Answer: b

b. **To read data from a file.**

Detailed Solution:

Input streams are used to read data from a file or another source.